

Thyagarajulu Gollapalli

Research Officer and Software Engineer, Australian National University

Melbourne, Australia gthyagi@gmail.com [gthyagi.github.io](https://github.com/gthyagi) [Google Scholar](#) [LinkedIn](#)

Research Interests

Computational geodynamics; geophysics; subduction dynamics; megathrust seismicity; mantle convection; fault representation in geodynamic models; numerical modelling; high-performance computing; reproducible open-source Earth-science workflows; geophysical data acquisition, processing, and interpretation.

Current and Past Work Experience

Research Officer/Software Engineer

Australian National University

Jan. 2026–Present

Canberra, ACT

- Open-source developer for the Underworld geodynamics codebase, contributing core features, user-facing interfaces, testing, documentation, and reproducible modelling workflows.

Assistant Lecturer, Maternity Cover

Monash University

Jun. 2025–Dec. 2025

Melbourne, VIC

- Taught geophysical interpretation, spatial analysis, modelling, machine learning, and deep-learning applications in geoscience and environmental science.

Senior Research Officer/Software Engineer

Monash University and AuScope

Aug. 2023–Jun. 2025

Melbourne, VIC

- Delivered research computing services for the AuScope Simulation, Analysis and Modelling node; developed Underworld tools, prepared documentation, and led training workshops.

Imaging Geophysicist

CGG

Nov. 2022–Jun. 2023

Perth, WA

- Applied seismic processing and subsurface imaging workflows, managed quality control, schedules, and HPC resources for geophysical imaging deliverables.

Junior Research Fellow; Project Assistant

IIT Bombay; Indian Institute of Science

Aug. 2015–Apr. 2018

Mumbai and Bengaluru, India

- Modelled lithospheric structure, rheology, and the Indian Ocean geoid low using gravity-geoid and computational geodynamics approaches.

Education

PhD, Computational Geodynamics and Geophysics

Monash University and Indian Institute of Technology Bombay

Apr. 2018–Nov. 2022

Thesis: *Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling*

Bachelor of Science, Mathematics; Minor in Earth and Environmental Science

Indian Institute of Science, Bengaluru, India

Aug. 2011–May 2015

Top Three Publications

1. Capitanio, F. A., **Gollapalli, T.**, Munukutla, R., Graciosa, J. C., Zuhair, M., Beall, A., & Dal Zilio, L. (2025). Bridging the gap between subduction dynamics and the long-term strength of megathrusts. *Nature Communications*. doi:10.1038/s41467-025-65824-7.
2. Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T.**, Lu, N., & Beucher, R. (2025). Underworld3: Mathematically self-describing modelling in Python for desktop, HPC and cloud. *Journal of Open Source Software*. doi:10.21105/joss.07831.
3. Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T.**, Tang, T., & Zuhair, M. (2025). Testing driving mechanisms of megathrust seismicity with explainable artificial intelligence. *JGR: Solid Earth*. doi:10.1029/2024JB028774.

Awards

- Faculty of Science Dean's International Postgraduate Research Scholarship, Monash University, 2018–2022.
- Monash Graduate Scholarship, Monash University, 2018–2021.
- Postgraduate Publications Award, Monash University, 2022.
- AGU Fall Meeting Student Travel Grant, 2021.
- Junior Research Fellow Scholarship, Ministry of Human Resource Development, India, 2017–2018.
- INSPIRE Scholarship for Higher Education, Department of Science and Technology, India, 2011–2015.